When Does a Speech Model Know to Hand Off? Reading Competence from Hidden States for Real-Time Escalation

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Abstract

A local speech model can keep a conversation going, and a stronger expert can take over the hard questions. Calling the expert on every turn adds cost and waiting time, so the local model needs an early estimate of whether its own answer will fail. Hidden-state probes give such estimates for text models. It is not known whether this signal survives speech and duplex fine-tuning, audio input, and the timing of a native listen/speak protocol. We study these questions on frozen MiniCPM-o 4.5. We use lightweight probes to read failure signals from hidden states and separately calibrate a gate for expert handoff at answer onset. We find that failure signals can transfer from text to speech, and that readout design matters in native audio generation. On our internal benchmark, selective expert handoff raises answer-content accuracy from 40.8% to 62.9%, while mean reconstructed time to first audio increases from 5.7 to 14.4 s. Serving speed therefore remains a key constraint for real-time use.

1 Introduction

Spoken interaction leaves little time to choose a response [Stivers et al., 2009]. One practical design pairs a local speech model, which keeps the conversation going, with a stronger expert that handles the hard questions. Calling the expert on every turn adds cost and waiting time. Selective escalation needs something else: an early estimate of whether the local answer will fail, made at the onset of a response.

Prior work gives two starting points. For text models, hidden-state probes and verbal self-evaluation can predict whether an answer is correct [Kadavath et al., 2022, Azaria and Mitchell, 2023, Kossen et al., 2024], and intermediate layers can carry information that the output does not show [Orgad et al., 2025]. Text routers use such signals to divide queries between models of different cost [Chen et al., 2023, Ong et al., 2024, Varshney et al., 2026]. For speech, native duplex models decide for themselves when to listen and when to speak [Défossez et al., 2024, Cui et al., 2026].

Three things remain unknown. First, it is not known whether a failure probe still works when the query pool changes; a high in-distribution AUC can come from pool identity rather than from competence. Second, it is not known whether a probe fit on text input still reads speech-input states after speech and duplex fine-tuning. Third, it is not known where in a streaming protocol the hidden state is actually available for a routing decision. A duplex model already decides when to speak, but this floor-management decision need not reveal whether it knows the answer. It is reasonable to expect that some competence information survives fine-tuning, because the backbone carried it before. The layer, position, and modality at which it can be read have not been measured on a duplex speech model.

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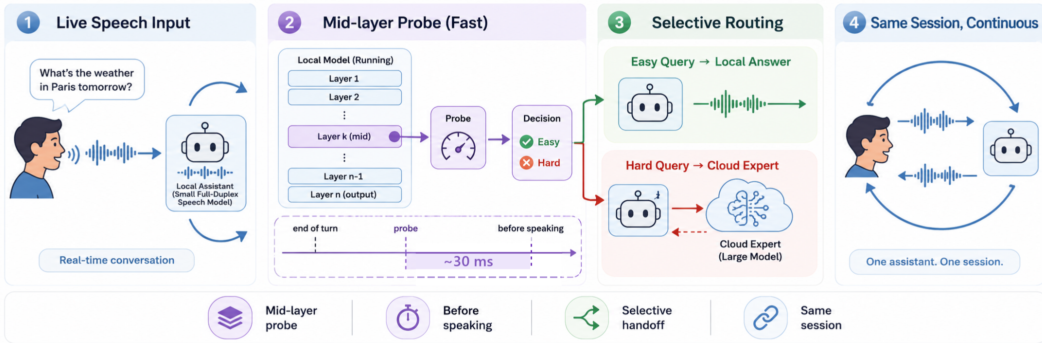


Figure 1: **The deployed system.** **(1)** Speech streams in 1 s chunks into a full-duplex local model that decides for itself when to speak. **(2)** At that speak commit, a mid-network (L22) linear read scores the turn in ∼30 ms at answer onset; the triggering call may already return the first text unit. **(3)** Lower-risk turns stay local; higher-risk turns are sent to a cloud expert while the talker stalls. **(4)** The returned answer is spoken verbatim in the talker's own voice. Delivered accuracy on the internal test set rises from 40.8% to 62.9% at 52% escalation in native full-duplex sessions.

This gap matters in practice. A gate placed at the wrong layer can look good on its calibration pool and then invert on a new one, so the expert is called on the wrong turns. A probe that fails on audio input cannot be used in a speech system at all. And a read that only becomes available after the answer is generated removes the early decision that motivated escalation in the first place.

We study these questions on frozen MiniCPM-o 4.5 [Cui et al., 2026], with raw-backbone and other speech-model controls. Our contributions are three. First, a layer and position audit shows that text-input LOPO math AUC increases from 0.372 at the final layer to 0.931 at layer 22. With audio calibration and audio input, the final-layer probe does not show this failure, so we use the middle layer for its robustness across the tested conditions, not because the output layer never holds competence information. Matched controls link the late-layer degradation to the tested fine-tuned checkpoints but do not identify its cause. At native answer onset, a matched 2×2 layer-by-readout ablation separates depth from aggregation: combining three positions raises L22 mean AUC from 0.595 to 0.695, while its single-state read trails the final layer (Figure 3). Second, a text-calibrated L22 probe transfers to speech-input states at AUC 0.855 on matched TTS queries and 0.760 on 200 query-disjoint human recordings (Figure 4). Cross-model comparisons extend the modality-transfer pattern to MiniCPM-o 2.6 and Qwen2.5-Omni, with weaker transfer in the frozen-backbone Freeze-Omni control. We separate this representation transfer from the extra calibration that native serving needs. Third, a native-protocol benchmark shows that a separately fitted answer-onset gate selects useful expert calls. It raises internal answer-content accuracy by 22.1 points and exceeds the reported random-mixture reference by 6.8 points at the aggressive tier. We report accuracy together with reconstructed time to first audio, and we distinguish this benchmark from a causal streaming evaluation of audible responses.

2 Related Work

Competence readout. Correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024], and uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]. Intermediate-layer representations can expose information missing from model outputs [Orgad et al., 2025, Mahaut et al., 2024]; Semantic Entropy Probes amortize uncertainty estimation into a hidden-state read [Kossen et al., 2024]. Our question is how readout location, input modality, and serving protocol affect this signal after speech and duplex fine-tuning.

Selective routing. FrugalGPT, HybridLLM, and RouteLLM allocate work across language models [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024]; AutoMix verifies a drafted answer before escalating [Aggarwal et al., 2024]. Pre-generation activation routers and frozen-state success probes are especially close to our method [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear failure probe is thus not itself the novelty. We test its robustness in speech-model representations and use a threshold to expose a tunable expert-call budget. Predicting local failure remains distinct from predicting the benefit of a particular expert.

Native duplex control. Native spoken-dialogue models coordinate listening and speaking [Défossez et al., 2024, Cui et al., 2026], using state or synchronization mechanisms [Wang et al., 2024a, Zhang et al., 2024, Veluri et al., 2024] or frozen-backbone adapters [Wang et al., 2024b]. BayLing-Duplex [Fang et al., 2026] makes dialogue-state decisions within a single autoregressive LLM by interleaving user audio, assistant text, and assistant audio and introducing four state tokens. Starting from GLM-4-Voice, it learns turn-taking and interruption through supervised full-duplex fine-tuning and timing-focused preference optimization. Its evaluation measures spoken-answer quality and interaction timing, providing a concrete example of learned native floor control.

Competence-aware handoff. DuplexOmni, chronological thinking, and MoshiRAG connect speech to reasoning or retrieval [Huang et al., 2026, Wu et al., 2026, Chien et al., 2026]. Our work studies a decision complementary to native floor control: given a duplex-trained checkpoint, can its hidden states predict local-answer failure and guide escalation to an expert under a tunable call budget? We fit a small readout without updating the speech model. This requires regime-specific calibration and does not assume that a learned turn-taking signal predicts answer correctness.

3 Probe-Guided Expert Handoff

3.1 The local–expert loop

Selective model routing balances response quality against the cost of a stronger model [Ong et al., 2024]. This suits our goal: improve answer accuracy under a limited expert-call budget. We adapt the routing decision to a local full-duplex speech model's transition from listening to speaking (Figure 1). The local model processes incoming speech and decides when to respond; the gate then chooses whether to continue locally or request expert help. On the expert path, an audio uplink transcribes the question, the expert returns a text answer, and the local talker voices the prepared relay text. The request and relay add cost and waiting time. We use local-answer failure risk as a proxy for the value of escalation: benefit still depends on whether the expert repairs the error and the relay preserves its answer.

3.2 A probe for the handoff decision

Linear *probes* test what information a classifier can read from frozen representations [Alain and Bengio, 2016]. Hidden-state classifiers have estimated statement truthfulness [Azaria and Mitchell, 2023] and pre-generation task success [Lugoloobi et al., 2026]. We use this approach to ask whether a speech model's existing states contain a directly readable failure signal. Reusing these states avoids generating a separate self-evaluation and leaves the speech model unchanged. The linear restriction makes this a test of accessible information; poor performance does not rule out information requiring a nonlinear readout.

Offline, we collect local answers to calibration questions and use a judge to assign correctness labels. Each label is paired with hidden-state features captured at the chosen read point. We fit a logistic probe to predict failure from these features while keeping the speech model frozen. For features $\phi(x)$ from input context x, the fitted weights w, b give the risk score

$$s(x) = \sigma\big(w^\top \phi(x) + b\big)\,. \tag{1}$$

At inference, the probe reads the current features without a reference answer or an online judge. Scores above a language-specific threshold τ_ℓ favor escalation; thresholds select nominal expert-call budgets from calibration scores and stay fixed during evaluation (§4.1). Realized call rates can change when the query distribution shifts. An auxiliary dialogue-act head restricts escalation to information requests, filtering turns such as acknowledgments and stop commands (Appendix D).

3.3 Where should the probe read?

Adapting the probe to duplex speech requires choosing its layer, position, and read time. The final layer feeds the output head, but this does not ensure that its failure signal survives a change in query distribution or input modality. We select L22 based on its robustness in the text-input layer audit and representation-transfer tests (§4.2–4.3); those results alone do not establish the best readout for native audio serving.

The native feature vector concatenates the latest L22 tail state, the mean of the tail states, and the mean input state, allowing the probe to combine the current boundary state with pooled context. The separate native layer-by-readout ablation (Figure 3) supports aggregation at L22; it does not show a mid-layer advantage for a single-token read.

We read these features when the generation call first returns a listen-to-speak transition. This makes the gate available at the local model's response boundary without waiting for a completed answer. The call can already contain the first generated text unit, so we call the decision point *answer onset*. Further answer chunks may expose additional failure information, at the cost of a later handoff (Appendix D.2). We fit the native probe and thresholds on features and answer labels from this serving regime to match its audio input and generation behavior. This calibration is separate from the representation-transfer study, which alone does not provide a zero-shot native gate.

4 Experiments

4.1 Experimental setup

We evaluate frozen MiniCPM-o 4.5 (9B), with `gpt-5.5` as the escalation expert, on the fixed 240-query internal test set and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], and SD-QA from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. The native probe is fitted on 5,221 calibration rows after excluding benchmark overlaps. All reported native-benchmark results were rerun with this refitted probe; per-language thresholds use the remaining core rows' out-of-fold scores and stay fixed across pools. Calibration details and model controls are in Appendix A.

Each query has one sampled native session per arm: local-only, conservative, balanced, aggressive, and always-escalate. We report *answer-content accuracy*, scored from generated local text or intended relay text. The random reference is the analytical mixture $A_{\mathrm{mix}}(r) = (1-r)A_{\mathrm{local}}+rA_{\mathrm{always}}$ at the gate's realized call rate r. Table 1 uses the original query weights; Figure 5's Internal curve uses the category weights specified in its caption. Both use the same recorded outcomes. External averages weight the four QA pools equally. The expert receives the final 30 seconds of the prerecorded question; timing is reconstructed from logged stages with different local and relay endpoints. These results therefore measure routing and content, without establishing causal streaming or client-audible latency. Full decoding, data, judging, and timing details are in Appendix A.1.

4.2 Layer and position matter

We first evaluate the read location independently of the expert path. A layer-by-position sweep under leave-one-pool-out (LOPO) transfer localizes the text-input failure to late last-token states (Figure 2). A last-token probe at L22 reaches **0.931** AUC on held-out math, compared with the final-layer baseline of 0.372, while mean pooling reduces the late-layer degradation. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96; the Qwen2.5 control family shows a similar ordering (Table 5). These comparisons associate the degradation with the tested fine-tuned checkpoints, without isolating a causal training objective.

With audio calibration and audio input, the final-layer probe reaches 0.936 on LOPO math. We retain L22 for robustness across the tested conditions; the mechanism behind the text-input degradation remains open (Appendix B.2).

At native answer onset, we separate the effect of layer from the effect of combining positions. All four readouts use one audio batch with identical training pools, labels, splits, and head recipe (Figure 3). At L22, aggregation raises mean failure-prediction AUC from 0.595 to 0.695; at the final layer, the change is smaller (0.654 to 0.662). The single-state L22 read has no advantage over the final layer. Thus, the evidence supports using aggregation with L22, rather than attributing the native signal to middle-layer selection alone. Named per-pool results and cached routing gains are reported in Appendix B.4.

Figure 2: **Text-input transfer depends on layer and position.** LOPO math AUC across layers: late last-token reads deteriorate on the duplex checkpoints, while mid-layer reads and raw-backbone controls remain predictive. Layer 22 is the MiniCPM-o 4.5 read location used for the native gate. This plot does not show an equivalent collapse under audio calibration and audio input.

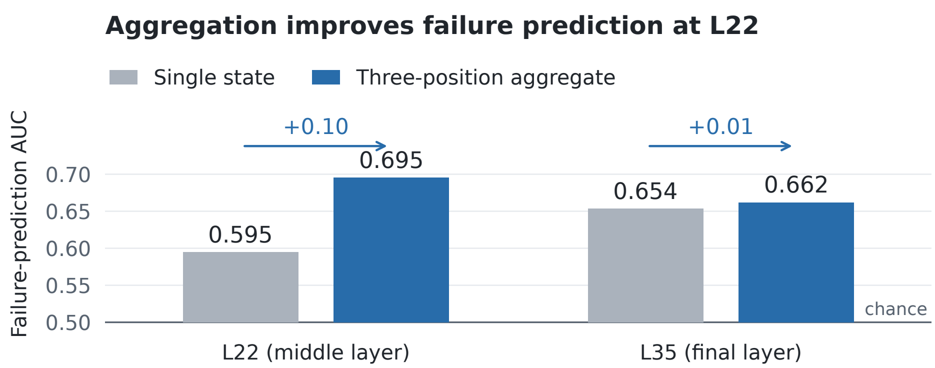


Figure 3: **Aggregation improves failure prediction at L22.** Mean AUC across five external pools under leave-one-pool-out evaluation on one native audio batch. Gray bars read the latest tail state; blue bars concatenate that state with the tail mean and input mean (§3.3). Arrows show the AUC change from aggregation; chance is 0.5.

4.3 Text-to-speech representation transfer

A text-calibrated L22 probe evaluated on speech-input states reaches **0.855** AUC. Figure 4 (left) locates where this readability emerges: probes trained on text-input hidden states and scored on speech-input states for the same queries are near chance in early layers and stabilize from roughly the middle of the network on every end-to-end-trained model — MiniCPM-o 4.5 (plateau 0.82–0.87), MiniCPM-o 2.6 (0.74–0.80, onset ∼46% depth), and the cross-family Qwen2.5-Omni control (0.80–0.83, onset ∼25% depth). The frozen-backbone Freeze-Omni adapter control never leaves the 0.52–0.60 band on identical backbone weights: the modality-shared mid-layer representation is created by training the backbone on the modality, not by the adapter alone. The two panels answer different questions and we keep them distinct: the left panel is *matched-query* modality transfer (same 600 queries as text and as TTS audio); the right panel is the strictly harder *query-disjoint* test — probes fitted on the calibration pool, evaluated on 200 held-out SD-QA human recordings — where the text-calibrated read on real speech reaches 0.760 at the deployed layer (band 0.76–0.80 over L22–L26), confirming the recipe is neither a TTS artifact nor memorization of the calibration questions (full per-layer tables in Appendix B.1). These results concern representation transfer. The escalation experiments below use a separately fitted native probe with three L22 positions, as defined in §3.3.

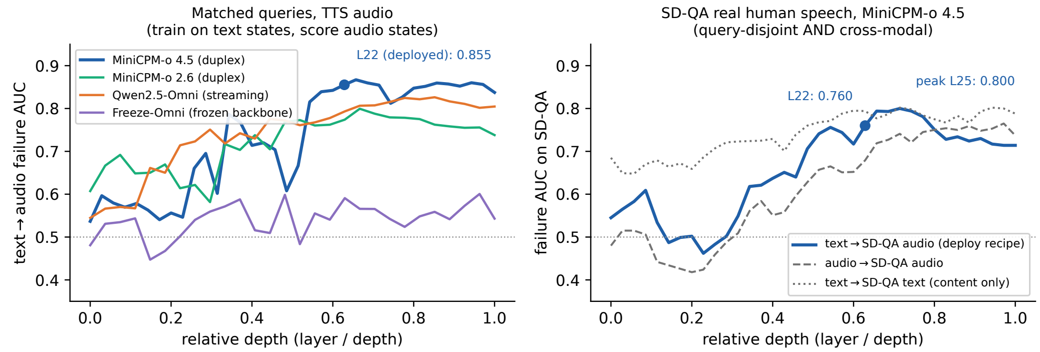


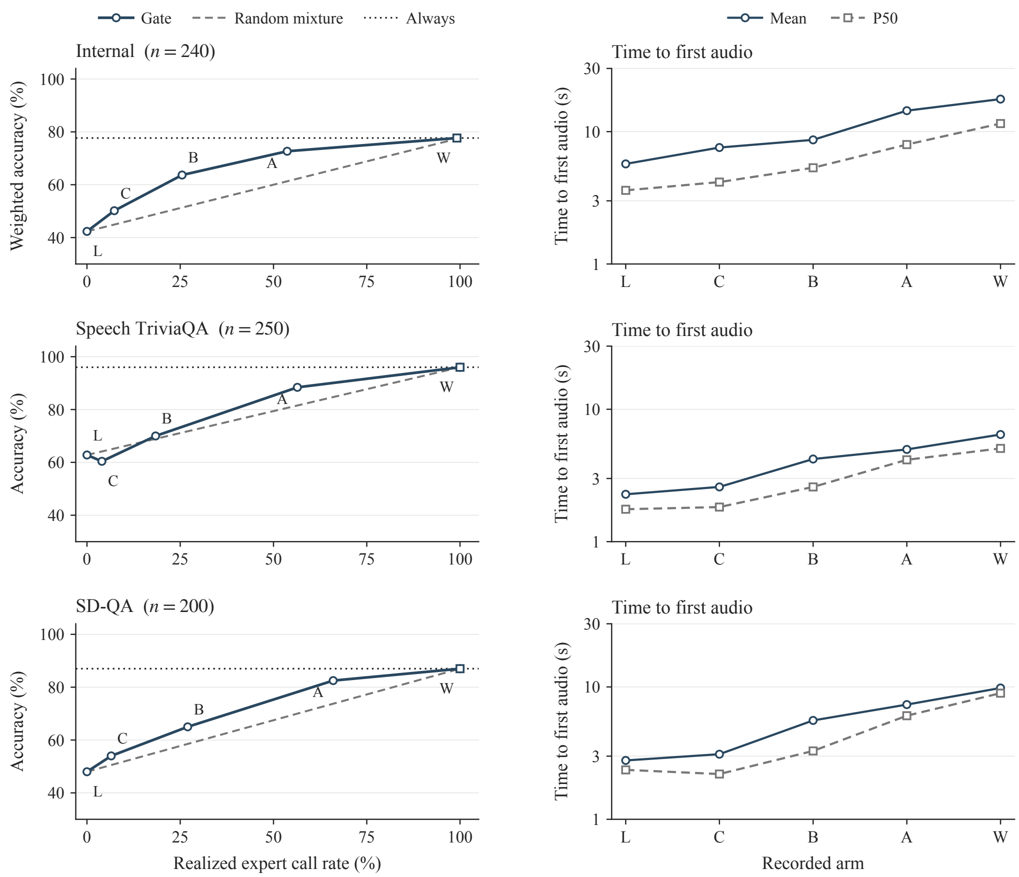
Figure 4: **Cross-modal readability emerges at mid-depth in end-to-end-trained models and transfers to real speech.** Left: per-layer failure AUC of probes trained on text-input hidden states and scored on speech-input states for the same queries (matched pairs, TTS audio). Both duplex MiniCPM generations and the cross-family Qwen2.5-Omni control develop a modality-shared mid-layer representation; the frozen-backbone Freeze-Omni adapter control does not. Right: the query-disjoint version on real human speech — calibration-pool probes evaluated on 200 held-out SD-QA recordings. The text-calibrated read (solid) tracks the audio-calibrated and text-target references and peaks in the same mid-to-late band (0.800 at L25; 0.760 at the deployed L22). Matched x-axes show depth relative to each model's layer count.

Table 1: **Main results: answer-content accuracy (%).** Both blocks report the internal test set and four external English speech-QA pools; avg is the equal-pool external mean and Δ is its gain over always-local, computed before rounding. MiniCPM uses one recorded native session per query and arm, with fixed calibration thresholds. Matched random mixes its local and measured always arms at each pool's realized gate rate (Appendix D.1). NVDA uses native-frame replay with cached outcomes; its Internal column is a post-selection diagnostic. Replay details and separate AlpacaEval scores are in Appendix A.1.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	avg	Δ
MiniCPM native sessions (answer-content accuracy)							
always-local	40.8	62.8	52.8	82.4	48.0	61.5	—
+ gate, conservative	46.7	60.4	59.2	78.8	54.0	63.1	+1.6
+ gate, balanced	54.2	70.0	62.4	82.4	65.0	70.0	+8.5
+ gate, aggressive	**62.9**	**88.4**	**75.2**	**85.6**	**82.5**	**82.9**	**+21.4**
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	77.1	+15.6
always-escalate (measured)	70.4	96.0	79.6	91.6	87.0	88.6	+27.1
NVDA three-layer answer-onset probe (pass-3 native-frame replay)							
always-local	19.6	35.8	37.6	70.8	31.0	43.8	—
+ gate @15%	31.7	48.4	48.8	80.7	42.0	55.0	+11.2
+ gate @30%	44.6	61.0	59.2	85.8	54.0	65.0	+21.2
+ gate @50%	60.4	76.8	69.6	89.7	66.5	75.7	+31.9
matched random @50%	53.5	65.7	60.2	82.0	60.0	67.0	+23.2
always-expert (ceiling)	91.7	95.5	82.8	93.1	89.0	90.1	+46.3

4.4 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports answer-content accuracy on the fixed 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive. Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix D.1). Aggressive exceeds the table's random reference of 56.1% by 6.8 percentage points. Per-tier internal comparisons are in Table 9. Aggressive recovers 74.6% of the measured local-to-always accuracy gap while using 51.7% expert calls. This measures selection value relative to the recorded endpoints.



Internal accuracy and call rate: weights of 0.25 for knowledge and math; 1 for other categories.
Timing uses the original query mixture.

Figure 5: **Accuracy gains and timing costs on three pools.** Left: answer-content accuracy against expert call rate. The Internal curve uses post-hoc weights of 0.25 for knowledge and math and 1 for other categories, for both accuracy and call rate; all 240 queries remain included. The Internal column of Table 1 retains its original query weights. External curves are unweighted. C/B/A denote conservative/balanced/aggressive. The dashed line uses the original Table 1 random-mixture convention (§4.1). The horizontal always-arm reference passes through the measured always point; it is not an oracle bound. Right: mean and P50 of the reconstructed time to first audio on a log scale, using the original query mixture. L/W denote local/always. Each point uses the original recorded pass, with one run per query and arm; no repeated-run uncertainty is implied. Local rows end at answer-text completion and escalated rows when the relay waveform is synthesized; playback duration is excluded (§4.1). Internal P95/P99 are reported in Table 10. All four external QA pools appear in the upper block of Table 1.

Gains vary across the fixed internal strata: factual QA improves by 52.5 points, while math accuracy is unchanged. Of 60 knowledge questions, 43 exceed the expert's input window. These descriptive results identify an input constraint and task differences, without attributing the accuracy gap to truncation or changing the test mixture (Appendix A.2).

The timing diagnostic makes the cost visible (Figure 5). Internal mean time to first audio increases from 5.7 s locally to 8.7 s at balanced and 14.4 s at aggressive; aggressive P95 is 60.8 s. The always arm averages 17.6 s (P50 11.5 s). The expert call itself accounts for 6.0 s of that mean; the rest is the blocking, non-streaming synthesis of the expert text, whose tail is driven by code and markup voiced verbatim (Appendix A). A lower call rate can reduce expert-path exposure, but these tails do not support calling the current aggressive setting low latency.

4.5 External transfer and its limits

The native gate's weights, feature definition, layer, and per-language thresholds stay fixed across the four external English QA pools. The upper block's *avg* excludes the internal set; Δ is its gain over always-local. Aggressive macro accuracy rises from **61.5% to 82.9%**, compared with 77.1% for the reported random mixture: +21.4 points over local answering and +5.9 over random. Its realized call rate ranges from 18.8% on Llama Questions to 66.0% on SD-QA, so this is not a uniform 50%-budget comparison. TriviaQA and SD-QA illustrate large improvements over local answering (+25.6 and +34.5 points) and over the corresponding random reference (+6.9 and +8.8 points).

The lower tiers are less reliable. Conservative accuracy falls by 2.4 points on TriviaQA and 3.6 on Llama Questions; balanced leaves Llama Questions unchanged. Separately sampled sessions and limited expert headroom constrain the interpretation of small differences. The results support aggressive-tier gains across these pools, not improvement on every benchmark at every budget. On AlpacaEval, we measure 3.68 locally and 4.01 at aggressive, with a random-mixture reference of approximately 3.9; the always arm scores 4.20 versus 4.95 for returned expert text (Appendix A.1). The relay cap is particularly restrictive for long, open-ended answers.

Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC increases from 0.755 at commitment to 0.791 after two answer chunks. This is consistent with some useful information emerging during generation, without proving a causal mechanism (Appendix D.2).

The lower block of Table 1 reports the exploratory NemotronLabs-VoiceChat-11B (NVDA) study. Its *avg* covers four English QA pools and excludes the Internal column. At a 50% replay budget, average accuracy rises from 43.8% to 75.7% ($\Delta = 31.9$ points), compared with 67.0% for random. This is native-frame replay with cached outcome mixtures, not another set of MiniCPM native sessions. The L26/L30/L34 heads reach 0.818 out-of-fold AUC and 0.816 mean external AUC in the separate probe analysis (Appendix C); Table 1 reports accuracy rather than AUC. External pools were inspected during variant review, so independent confirmation is needed.

5 Discussion and Limitations

The central finding is that a mid-layer read remains predictive in tested conditions where a final-layer, last-token probe fails. This read location also supports selective expert calls without updating the speech model.

Three boundaries define the scope of the results. First, coverage is limited to the tested checkpoints and predominantly single-turn speech QA. Broader conversational tasks, including multi-turn dialogue and expert continuity, remain to be evaluated. Second, native serving requires in-regime audio features and answer labels to fit the readout and set thresholds; representation transfer alone does not provide a zero-shot gate. Third, the benchmark measures answer-content accuracy and reconstructed serving time. Its long expert-path tails do not establish a latency benefit; human-rated speech quality and common-clock measurements of audible responses would better capture the listener's experience.

These results motivate competence-aware handoff as a useful addition to a speech model's floor control. Appendix A records the calibration, scoring, and input protocol needed to interpret the benchmark.

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A Benchmark and calibration details

The main table reports the original internal query mixture and four external English QA pools. The figure's Internal accuracy and call-rate curve uses post-hoc knowledge/math weights of 0.25 and weights of 1 for other categories; timing retains the original mixture. Archived Mandarin diagnostics are separate from the main-table macro average.

A.1 Experimental setup details

Models, data, and calibration. The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Matched controls include raw backbones, a second MiniCPM generation, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni (Table 3). Representation analyses use bf16, greedy decoding, and seed 42. The native benchmark uses sampled decoding under the serving configuration ($T = 0.7$, top-$k = 20$); these are different experimental regimes. The escalation expert is `gpt-5.5` with low reasoning effort.

The internal 600-query set contains SimpleQA traps [Wei et al., 2024], MMLU-Pro knowledge questions [Wang et al., 2024c], TriviaQA facts [Joshi et al., 2017], Dolly/Alpaca chat prompts [Conover et al., 2023, Taori et al., 2023], and GSM8K/MATH problems [Cobbe et al., 2021, Hendrycks et al., 2021, Lightman et al., 2024]. Its original 360/240 split and test composition remain fixed. LLM judges score answer adequacy. A stronger re-judge of the representation-study labels agrees on 96.2% of text and 94.5% of audio cases; this agreement is not a validation of speech delivery.

Three calibration regimes must be distinguished: 360 examples for single-position representation probes; 2,310 for the earlier turn-based three-position gate; and 5,221 for the native gate, refitted on native features and judged native-answer outcomes. The native fit contains 4,979 core and 242 fresh training rows; thresholds use five-fold out-of-fold scores from the core rows, grouped by language. Conservative, balanced, and aggressive target nominal calibration rates of 15/30/50%; realized rates on new pools can differ. Neither fresh-row scores nor external evaluation-pool quantiles enter these thresholds. Seven rows overlapping external questions were removed from the calibration set before refitting the gate and recomputing its thresholds. All native-benchmark results reported here come from reruns with this deduplicated fit (Appendix A).

Native benchmark protocol and metrics. The MiniCPM upper block of Table 1 evaluates the internal test split and four external English speech-QA pools: Speech TriviaQA, Speech Web Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], plus SD-QA recordings from VoiceBench [Faisal et al., 2021, Chen et al., 2024]. VoiceBench AlpacaEval is reported separately on its 1–5 scale. The archived Mandarin Speech Reasoning QA pool is retained in the appendix diagnostics and excluded from the main-table macro average. External audio comes from the benchmark releases; internal questions use a fixed TTS rendering. Each query has a separate native-protocol session for each arm: local-only, three gate tiers, and always-escalate. There is one recorded run per query and arm, without repeated-run averaging.

Audio enters the local model in one-second chunks. At an escalation, transcription and the expert produce a text answer, which is cleaned and capped at 400 characters for teacher-forced TTS relay. We report *answer-content accuracy*: the judge sees local generated text or the intended relay text, not a transcription of played audio. The benchmark disables local audio generation. Uplink transcription uses the final 30 seconds of the full prerecorded question: it can access unconsumed audio at early onset and loses the beginning of longer questions. Thus the experiment evaluates routing under this input protocol, without establishing strictly causal interaction.

For the MiniCPM block, the reported random reference is

$$A_{\text{mix}}(r) = (1 - r)A_{\text{local}} + rA_{\text{always}},$$

where r is the gate's realized escalation rate. This follows Table 1's convention of treating the always policy as the full-escalation endpoint; it is an analytical mixture, not a separately run randomized arm. Non-commits and a sensitivity check using the actual always-arm rate are reported in Appendix A. Timing is reconstructed from chunk offsets and logged stage durations as time to first audio: the relay path ends when its synthesized waveform is ready (expert wait plus blocking TTS), excluding the waveform's playback duration; the local path ends at text generation. Figure 5 reports mean and P50 of this diagnostic; Table 10 retains internal P95/P99. These summaries do not measure client-audible completion (Appendix A).

Main-table replay and open-ended scores. The NVDA lower block re-mixes local and cached expert outcomes after pass-3 replay through the native frame protocol (Appendix C). External columns retain cached local outcomes and the prior 2,481-row fit. The Internal column judges the same pass-3 answers on all 240 queries and uses a 2,258-row refit that excludes these test IDs, with cached pass-2 training labels. All policies leave the 17 queries without an answer-onset read local, so 15/30/50% scoreable-query budgets realize 14/28/47% overall. This Internal result is post-selection because probe architecture selection included this split.

AlpacaEval uses a separate 1–5 scale: MiniCPM changes from 3.68 locally to 4.01 at 44% escalation (rounded-anchor matched random 3.91; always-escalate 4.20, compared with 4.95 for returned expert text). The pass-3 NVDA ranking changes from 3.55 to 4.51 versus 4.23 for random. These open-ended scores do not enter the QA macro average.

Calibration. Representation analyses use a fixed 360-query calibration split and single-position 4,096-dimensional features. The native gate uses 5,221 rows with native-answer labels and three L22 summaries: the last tail state, the tail mean, and the input mean. The logistic head uses $C = 3 \times 10^{-4}$. Five-fold stratified cross-validation uses seed 42; thresholds are language-specific 15/30/50% quantiles of the 4,979 core rows' out-of-fold scores. The 242 fresh rows enter fitting but not the quantile base. Exported scores reproduce thresholds to four decimals. The overlap audit identified seven calibration rows matching external questions. These rows were excluded from both fitting and threshold calibration, and the native benchmark was rerun with the resulting gate. All reported native-benchmark results use this deduplicated fit. Layer selection used internal LOPO results without nested selection within each held-out pool.

Scoring and timing. Each arm contains one native-protocol session per query. Answer-content accuracy is judged from local generated text or the intended expert relay text. The expert receives the last 30 seconds of the full question waveform. Of 240 internal questions, 46 exceed that window; the always arm escalates 45 of these and drops their beginnings. The input includes not-yet-consumed audio for 18/240 aggressive sessions. Time to first audio is reconstructed from the onset chunk offset and generation, waiting, and synthesis times; relay playback duration is excluded. Local rows end at text generation, while escalated rows end when blocking TTS returns a waveform. The always arm fires on 238/240 queries. Adjusting the random reference for its 99.2% call rate changes the aggressive reference from 56.1 to 56.2%, leaving a 6.7-point gate advantage.

Repeated and audio-output checks. Two additional internal passes put aggressive accuracy in the range 62.9–65.4% (mean 64.4%, sample standard deviation 1.3 points). A separate 299-query validation uses only consumed input audio and saves both output waveforms. Local/always text accuracy is 50.8/77.3%, while accuracy after output-audio transcription is 48.2/66.2%. The validation uses a separate 4,922-row gate fit excluding its 299 queries and reports endpoint arms, rather than live gate-arm results. The repeats describe internal run variability; the transcription scores motivate direct speech evaluation.

A.2 Internal task strata

Table 2: Descriptive strata of the fixed internal test set. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm's realized rate. The final column counts question waveforms longer than the expert's 30-second input window. One recorded run per query and arm; no samples are removed or reweighted.

Stratum	n	Local	A	Always	A call rate	> 30 s (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The largest gains occur on factual QA and SimpleQA traps. Math is unchanged at the aggregate level, consistent with the failure-type analysis in Appendix D.2. Long-input truncation is concentrated in the knowledge stratum (43 of 60 questions); all queries remain in the reported denominator.

B Models and representation analyses

Table 3: Model roster. "Pair" identifies a fine-tuned model and its raw backbone control.

model	type	pair	role
MiniCPM-o 4.5	omni + duplex FT	Qwen3-8B	system target
Qwen3-8B	raw	—	pair control
MiniCPM-o 2.6	duplex FT	Qwen2.5-7B	replication
Qwen2.5-Omni-7B	streaming omni FT	Qwen2.5-7B	omni control
Qwen2.5-7B	raw	—	pair control
Qwen2-Audio-7B	audio FT	Qwen2-7B	audio control
Freeze-Omni	frozen LLM + adapter	Qwen2-7B	adapter control
Qwen2-7B	raw	—	pair control

B.1 Signal comparison and layer transfer

Table 4: Failure-prediction AUC on MiniCPM-o 4.5. In-dist uses five-fold out-of-fold predictions; LOPO holds out the named pool.

signal	in-dist	LOPO math	audio input
max entropy@4	0.696	near chance	—
$p(\text{True})$-pre	0.807	trap 0.945	degrades
$p(\text{True})$-post	**0.899**	—	degrades
final-layer probe	0.822	**0.372**	0.936
mid-layer (L22) probe	0.866	**0.931**	**0.855**

The final-layer state predicts source pool at 95.8% accuracy, and a pool-identity oracle alone reaches failure AUC 0.715. This motivates LOPO rather than an in-distribution comparison. The depth sweep shows that the failure is localized: mid-layer last-token reads transfer while late reads degrade on both duplex models; mean pooling degrades less. In the signal table, the final-layer audio entry is audio-calibrated LOPO math, whereas the L22 audio entry tests a text-calibrated probe on speech states.

Table 5: LOPO math AUC by depth for matched model families.

model	best mid-layer	final layer	shape
Qwen3-8B	0.964 (L33/36)	0.958	plateau
MiniCPM-o 4.5	**0.931 (L22/36)**	**0.366**	late inversion
Qwen2.5-7B	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni	0.794 (L16/28)	0.746	diffuse decline
MiniCPM-o 2.6	**0.822 (L21/28)**	**0.540**	late cliff

The text-to-audio result is not explained by synthetic speech alone: on 200 human SD-QA recordings, the text-calibrated read remains at 0.76–0.80 AUC across L22–L26. Audio-input verbal self-evaluation is less reliable, whereas adding a transcript restores it; the hidden-state probe does not require that extra generation step.

B.2 Mechanism checks

Matched controls narrow the late-layer effect without assigning it to a specific training objective. A speak-mode template leaves final-layer LOPO math unchanged (0.362 versus 0.366), the audio–text direction is nearly orthogonal to the failing probe, and source-pool identity is decodable at all depths. A residual-component analysis shows the distributed read holding through the middle layers and decaying only near the outputs of the two duplex models; it remains intact in their raw backbones and the omni-streaming control. This supports the practical choice of a mid-layer read while leaving the causal mechanism open.

B.3 Query-surface baseline

A TF–IDF logistic router reaches 0.669 out-of-fold AUC and 0.38–0.57 under LOPO, including 0.230 on the all-failure trap pool. The hidden-state probe's advantage is therefore not explained by query wording or source identity alone. On audio labels the surface router improves to 0.814 but still trails the L22 probe at 0.879.

B.4 Native feature ablation

Figure 3 uses one native audio batch (`agg0`, official configuration), with all layer features captured in the same sessions and judged against their own generated answers. Single-state and aggregated features have 4,096 and 12,288 dimensions, respectively. Each readout fits a logistic head ($C = 3 \times 10^{-4}$) on four pools and evaluates the fifth. The five pools include Mandarin reasoning; this set differs from the main table's four English external pools. Table 6 identifies the individual pool results that underlie the averages, including the weaker reasoning scores.

The routing comparison selects the highest-risk 30% of questions in each pool and combines recorded local outcomes with cached always-arm expert outcomes. Figure 6 reports pooled accuracy gains over 2,000 matched random allocations. Per-pool gains in Table 6 use the analytical 30% random mixture. These are cached outcome comparisons, not new live gate sessions. For the L22 single-state read, 0.0505 of the random allocations matched or exceeded the observed accuracy; none did for the other three readouts. These comparisons test against random routing, not differences between the readouts.

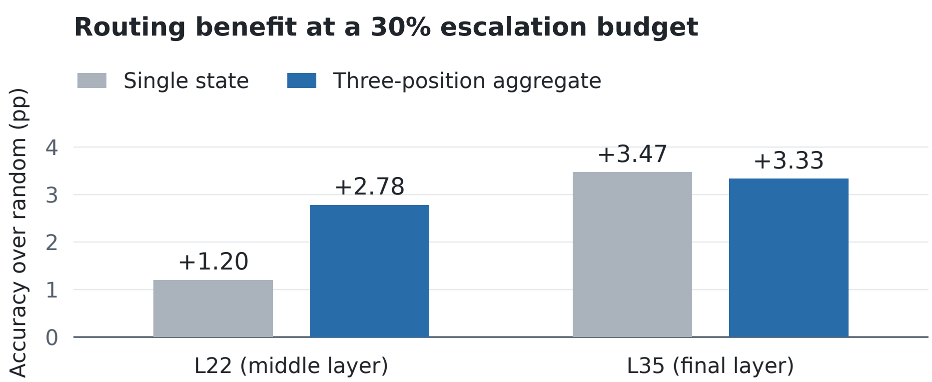

Figure 6: **Cached routing gains at a matched 30% budget.** Accuracy gains over random escalation, pooled across the same five evaluation sets as Figure 3. Each bar uses the recorded local and expert outcomes; feature choices and colors match the main figure.

For context, the separately trained deployed gate reaches mean AUC 0.750 and a pooled gain of 4.80 percentage points on these rows. It was fitted on the full calibration merge, so it is not a matched training condition for the four LOPO ablations and is excluded from their plots.

C Second duplex family

We replay NemotronLabs-VoiceChat-11B through its native frame protocol with the checkpoint frozen. Of 2,550 English calibration queries, 2,481 produce a commit frame. Standardized logistic heads read L26/L30/L34 onset states; their normalized logits are averaged. The recipe is chosen on calibration-only LOPO metrics, and external labels are excluded from fitting. Each head uses $C = 10^{-4}$ and concatenates the commit state, eighth post-commit state, first-eight-state mean, and running pre-commit mean (17,920 dimensions per layer). External pools were inspected during variant review, so these results provide exploratory transfer evidence. The read covers the first 640 ms after commit, so this experiment tests method transfer rather than end-to-end real-time serving.

Table 6: Native ablation by evaluation pool. Each column is one of the four matched LOPO readouts in Figure 3. The lower block gives accuracy gains over random escalation in percentage points. AUC averages weight pools equally; pooled routing gains weight individual questions.

	L22		L35	
Pool	Single	Aggregate	Single	Aggregate
Failure-prediction AUC				
Speech TriviaQA	0.645	0.744	0.704	0.661
Speech Web Questions	0.579	0.732	0.646	0.683
Speech Llama Questions	0.614	0.711	0.672	0.674
SD-QA	0.648	0.713	0.682	0.669
Reasoning QA (zh)	0.488	0.576	0.564	0.624
Equal-pool mean	0.595	0.695	0.654	0.662
Accuracy gain over random at a 30% budget (pp)				
Speech TriviaQA	+3.38	+4.58	+3.78	+3.38
Speech Web Questions	+1.69	+1.69	+4.16	+3.33
Speech Llama Questions	+1.51	+3.96	+2.73	+3.96
SD-QA	+2.85	+5.35	+5.85	+4.35
Reasoning QA (zh)	-3.91	-1.93	+1.04	+1.53
Pooled gain	+1.20	+2.78	+3.47	+3.33

Table 7: VoiceChat-11B transfer. Mean accuracy equally weights four English speech-QA pools; random is the matched-rate mixture.

expert budget	15%	30%	50%
probe-guided accuracy	55.0	65.0	75.7
random-mixture accuracy	50.7	57.7	67.0

The three-layer read reaches 0.818 out-of-fold AUC and 0.816 mean external AUC under the pool-aligned judges. A strictly causal commit-only variant loses about 0.03 external AUC, making answer onset the closest matching read point across the two model families.

D Native gate and serving details

The L22 hidden state is available after roughly 57–61% of prefill. On one H100, probe availability has P50/P95 of 20/25 ms for text and 45/104 ms for audio. These numbers measure the local read, not the network or expert call. The native benchmark's realized rates are below; their variation is expected because thresholds are calibration quantiles rather than per-pool quotas.

Dialogue-act filtering. A second linear head uses the same L22 read to distinguish information requests from floor-management turns. Escalation requires both the information-seeking score $a(x)$ and failure score $s(x)$ to exceed their respective thresholds:

$$d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\}\,\mathbf{1}\{a(x) \geq \tau_a\}. \tag{2}$$

On the scripted calibration stimuli, the act head has OOF AUC 1.000; the question-side 0.5th-percentile threshold removes 0.5% of true escalations and admits no floor-management stimuli.

D.1 Realized escalation rates

Table 8: Realized escalation rates (%) for the MiniCPM native results in the upper block of Table 1, plus the archived Mandarin and separate AlpacaEval pools. These are measured rates, distinct from the nominal calibration budgets and from the NVDA replay budgets. Local-only makes no expert calls.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

Table 9: Internal native-protocol answer-content accuracy with query-bootstrap 95% confidence intervals ($n = 240$). Random follows Table 1's endpoint-mixture convention.

tier	realized calls	accuracy	random reference
local-only	0.0%	.408 [.35,.47]	.408
conservative	8.3%	.467 [.40,.53]	.433
balanced	25.0%	.542 [.48,.60]	.482
aggressive	51.7%	.629 [.57,.69]	.561
always	99.2%	.704 [.65,.76]	.704

Table 10: Internal reconstructed time to first audio (seconds, $n = 240$ per arm). Local rows end at text completion and escalated rows when blocking TTS returns. Playback duration is excluded. Quantiles describe variation across queries within one run.

tier	mean	P50	P95	P99
local-only	5.7	3.6	17.3	21.2
conservative	7.6	4.2	17.5	80.2
balanced	8.7	5.3	22.9	89.6
aggressive	14.4	8.0	60.8	131
always	17.6	11.5	69.4	117

D.2 Failure type and later reads

This separate sweep fits a probe at each read point using 5,236 judged calibration rows, row-random five-fold fitting, and $C = 3 \times 10^{-4}$. Later reads pad finished answers with the last available vector. Its five-pool macro includes Mandarin reasoning, unlike the main table's four-English-pool macro. Failure categories are model judgments.

Failure-prediction AUC at successive native read points. External macro equally weights five speech-QA pools.

Read point	Calibration OOF	Internal	Reasoning-zh	External macro
Before onset	.848	.832	.711	.753
At onset	.850	.852	.648	.755
After 1 chunk	.852	.854	.681	.762
After 2 chunks	.855	.846	.769	.791
After 3 chunks	.857	.858	.744	.801

Specific-but-wrong answers account for about 70% of failures and are ranked at AUC 0.813; perception errors reach 0.889, while execution slips are near chance at 0.469. The later-read gains on reasoning expose a tradeoff between earlier routing and information that emerges during generation.

E Judge prompt and reproducibility

The answer judge receives the question, an optional reference, and a candidate answer without the system identity. It marks the candidate adequate only when it is factually correct, uses valid reasoning, and completely answers the question; partial, hedged, refused, off-topic, and truncated answers are inadequate. The pre-answer self-evaluation asks the model not to answer the question and to reply exactly “Yes” or “No” to whether it would answer correctly. The post-draft version presents the question and proposed answer and asks whether the answer is correct.

Representation-study sampling and splits use seed 42. The native benchmark uses sampled decoding and retains one run per query and arm. MiniCPM-o 4.5 runs with PyTorch 2.8.0 and Transformers 4.51.0 on single-H100 instances. The release includes the calibration manifest, out-of-fold scores, thresholds, overlap exclusions, benchmark hashes, and scripts that rebuild the figures and metric summaries from judged rows.